
PH 515P: Physics Laboratory

Course Code	PH 515P
Course Name	Physics Laboratory
L-T-P-C	0-0-5-3
Prerequisites	No
Intended for	I-Ph.D. and M. Sc Physics students
Mutual Exclusion	NA
Approval	10th Senate (Jun 2016)
Revised	64th BoA (May 2026) - syllabus revised

Preamble

This experimental course is expected to develop the art of experimentation and analysis skill, understanding the basis of knowledge in physics, and collaborative learning skills among students. The course content includes standard physics experiments from various modules of physics, the theory of which students have learnt during their final year of B. Sc.

Course Modules

No

Laboratory Modules

1. Hall Effect in Semiconductor
Objective: To measure the resistivity and Hall voltage of a semiconductor sample as a function of temperature and magnetic field. The band gap, the specific conductivity, the type of charge carrier and the mobility of the charge carriers can be determined from the measurements.
2. Michelson Interferometer
Objective: To determine the wavelength of the light source by producing interference pattern.
3. Fabry-Perot Interferometer
Objective: To investigate the multibeam interference of a laser light. Also, the wavelength of light source and thickness of a transparent foil can be determined.
4. Zeeman Effect
Objective: To observe the splitting up of the spectral lines of atoms within a magnetic field (normal and anomalous Zeeman effect) and find the value of Bohr's magneton.
5. Diffraction of ultrasonic waves
Objective: To observe Fraunhofer and Fresnel diffraction and determine the wavelength of the ultrasound wave.
6. Frank-Hertz Experiment
Objective: To demonstrate the quantization of atomic energy states and determine the first excitation energy of neon.
7. Fourier optics
Objective: To observe Fourier transformation of the electric field distribution of light in a specific plan.

8. Dispersion and resolving power of a grating
Objective: Determination of the grating constant of a Rowland grating based on the diffraction angle (up to the third order) of the high intensity spectral lines. Determination of the angular dispersion and resolving power of a grating.
9. Geiger-Müller-Counter
Objective: To study random events. Determination of the half-life and radioactive equilibrium. Verification of the inverse-square law for beta and gamma radiation.
10. Scintillation counter
Objective: Energy dependence of the gamma absorption coefficient / Gamma spectroscopy.
11. Four Probe
Objective: To measure the resistivity of a semiconductor at different temperature, finding the band gap.
12. Millikan Oil drop
Objective: To estimate the electrical charge of an electron and also show that charge is quantised in units of e .
13. Magnetoresistance of a semiconductor
Objective: To study the magnetic field dependence of the transverse magnetoresistance of a semiconductor.
14. Thermal conductivity of Teflon
Objective: To determine the thermal diffusivity coefficient of Teflon (an organic solid material) using (a) linear flow of heat and (b) radial flow of heat and estimate its thermal conductivity.

Text Books

1. R. A. Dunlop, *Experimental Physics*, Oxford University Press (1988).
2. J Varma, *Nuclear Physics Experiments*, New Age Publishers (2001)

References

1. A. C. Melissinos, *Experiments in Modern Physics*, Academic Press (1996).
2. E. Hecht, *Optics*, Addison-Wesley; 4 edition (2001)
3. Worsnop and Flint, *Advanced Practical Physics for Students* Methusen & Go. (1950).
4. E.V. Smith, *Manual for Experiments in Applied Physics*. Butterworths (1970).
5. D. Malacara (ed), *Methods of Experimental Physics*, Series of Volumes, Academic Press Inc. (1988).